

2021-24

Full Marks : 60

Time : 3 Hours

Candidates are required to give their answer in their own words as far as practicable. Their figures in the margin indicate full marks.

Answer from **both** the Sections as directed.

Section – A/(Compulsory)

1. Select the most appropriate option of the following statement : $1 \times 10 = 10$

(i) Which of the following is an objective of network security :

- (a) Confidentiality (b) Integrity
(c) Availability (d) All of the above

(ii) Identify the term which denotes that only authorized users are capable of accessing the information :

- (a) Confidentiality (b) Integrity
(c) Availability (d) All of above

(iii) From the options below, which of them is **not** a vulnerability to information security :

(a) Flood

(b) Without deleting data, disposal of storage media

(c) Unchanged default password

(d) Latest patches and updates not done

(iv) Possible threat to any information cannot be :

(a) Reduced

(b) Transferred

(c) Protected

(d) Ignored

(v) A firewall protects which of the following attacks ?

(a) Phishing

(b) Dumpster diving

(c) Denial of Service (DoS)

(d) Shoulder surfing

(vi) What is used for database security ?

(a) A view

(b) Data Encryption

(c) Finger print

(d) All of the above

(vii) The addition of multilevel security to a database system does not increase the complexity of the access control function.

- (a) True (b) False

(viii) Salami attacks is a type of attack that operates on :

- (a) Confidential data
(b) Company Data
(c) Financial Data
(d) Employee Data

(ix) In cryptography, what is cipher :

- (a) Algorithm for performing encryption and decryption.
(b) Encrypted message.
(c) Both algorithm for performing encryption and decryption and encrypted message.
(d) Decrypted message.

(x) The legal risks of ethical hacking include lawsuits due to of personal data :

- (a) Stealing (b) Disclosure
(c) Deleting (d) Hacking

2. What is firewall ? What is the usage of it ?

5

Section - B

Answer any **three** questions : $15 \times 3 = 45$

3. (i) What is the difference between encryption and hashing ?
(ii) What is the difference between symmetric and asymmetric encryption.
4. (i) What are the security services of Information security ?
(ii) What is attack in information security ? What are the types of attacks ?
5. (i) What is database security ? Explain requirements of database security.
(ii) Explain the term reliability and Integrity in database security with example.
6. (i) What is trap door and salami attack ? Explain with example.
(ii) Explain ethical issues in Information security.
7. Write short notes on (any **two**) : $7.5 \times 2 = 15$
 - (i) Intrusion Detection System.
 - (ii) Digital Signature.
 - (iii) File Protection.
